Corrected and conduct of chapter 10

Use as a teacher. The following answers relate to the outstanding tasks in the course. first clarify the approach; provide graduated help before the solution. estimates to be calibrated in class.

Diagnosis and Remediation

1. 2π. *Action:* Connect radius, diameter and perimeter.
2. 90°. *Action:* Share a circle in four sectors.
3. $\dfrac{\sqrt{2}}{2}$. *Action:* Streamline and check square.
4. $(a;-b)$. *Action:* Draw both points.
5. $-\dfrac{\pi}{3}$. *Action:* Put at the same denominator.
6. $\dfrac{3}{5}$. *Action:* Identify the reference angle.

Research activity

Equilateral triangle then isosceles right triangle. Pythagoras gives respectively height $\dfrac{\sqrt{3}}{2}$ and equal coordinates $\dfrac{\sqrt{2}}{2}$. The signs are positive in the first quadrant.

Immediate applications

1. $\dfrac{\pi}{4}$, $\dfrac{2\pi}{3}$; length 3 $\dfrac{\pi}{2}$ m.
2. $\dfrac{7\pi}{2}$ maps to the value $\dfrac{3\pi}{2}$; $-\dfrac{5\pi}{3}$ that of $\dfrac{\pi}{3}$.
3. $\sin^2 x = \dfrac{16}{25}$; the sine is negative, so $-\dfrac{4}{5}$.
4. $(-\dfrac{1}{2};-\dfrac{\sqrt{3}}{2})$ and $(\dfrac{\sqrt{2}}{2};-\dfrac{\sqrt{2}}{2})$.
5. $h = \dfrac{\sqrt{3}}{2}$; new rope $\dfrac{1}{\sqrt{2+\sqrt{3}}} = \sqrt{2-\sqrt{3}}$.

Exercises: approach and results

Exercise 10.1 · Convert and measure. $\dfrac{5\pi}{6}$ and $-\dfrac{\pi}{2}$; arc $\dfrac{4\pi}{3}$ cm. The negative sign indicates a orientation; geometric length is positive.

Exercise 10.2 · Reduce angles. $\dfrac{5\pi}{6}$, $\dfrac{\pi}{4}$, π. Remove or add multiples of 2 π.

Exercise 10.3 · Accurate values. $(-\frac{1}{2};\frac{\sqrt{3}}{2})$; $(\frac{\sqrt{3}}{2};-\frac{1}{2})$; $(-\frac{\sqrt{2}}{2};-\frac{\sqrt{2}}{2})$ in form (cosine ; sine).

Exercise 10.4 · Find the sign. $\cos^2 x = 1-\frac{25}{169}=\frac{144}{169}$; negative cosine, therefore $-\frac{12}{13}$.

Exercise 10.5 · One wheel. Progress $r\theta = 0{,}3(\frac{5\pi}{3})=\frac{\pi}{2}$ m. This corresponds to the course of the arc; the rope connects two positions on the circle and does not have that length. The material point on the wheel describes a different trajectory from the ground.

Exercise 10.6 · String and scalar product. $A(1;0)$, $B(-\frac{1}{2};\frac{\sqrt{3}}{2})$; $AB^2=\frac{9}{4}+\frac{3}{4}$ = 3. Al-Kashi : $1+1-2\cos(\frac{2\pi}{3})$ = 3. $AB=\sqrt{3}$.

Exercise 10.7 · Read an equation on the circle. The vertical abscisse $\frac{1}{2}$ cuts the circle to the angles $\frac{\pi}{3}$ and $\frac{5\pi}{3}$. Both points have opposite sinuses.

Exercise 10.8 · Deferred review. $\frac{25\pi}{4}=6\pi+\frac{\pi}{4}$. Three full turns do not change the image, So the two values are worth $\frac{\sqrt{2}}{2}$.

CQFD, exercise 32 p. 220, questions a and b. a : $\cos(\frac{6\pi}{5})=\cos(\pi+\frac{\pi}{5})=-\frac{1+\sqrt{5}}{4}$. b : $\cos(-\frac{\pi}{5})=\cos(\frac{\pi}{5})=\frac{1+\sqrt{5}}{4}$, by cosine parity.

CQFD, Exercise 39 p. 220, Questions 1, 2, 3 and 5. $1:\frac{\pi}{6}+\frac{\pi}{3}=\frac{\pi}{2}$. 2 : yes, $\cos(\frac{\pi}{3})=\frac{1}{2}=0{,}5$ is decimal. 3: the main measure in] $-\pi;\pi]$ is $-\frac{5\pi}{6}$. 5 : $\sin(\frac{4\pi}{3})=-\frac{\sqrt{3}}{2}$.

MCQ: answers and reasons

MCQ 1.

1. B. Half-turn.

2. A. Equilateral triangle.

3. D. Lower point of the circle.

4. C. Equation of the unity circle.

5. B. Complete towers.

MCQ 2.

1. C. Vertical Symmetry.

2. A. Isosceles right triangle.

3. D. Three-quarters of a turn.

4. B. $r\theta = 6$.

5. C. Negative abscess inside the quadrant.

True or false

1. False: in 0, the two members would be 1 and -1. Cosine is even.

2. True: a complete tower preserves the image.

3. False: $\frac{\pi}{6}$ and $5\frac{\pi}{6}$ have even sine but cosine opposites.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.